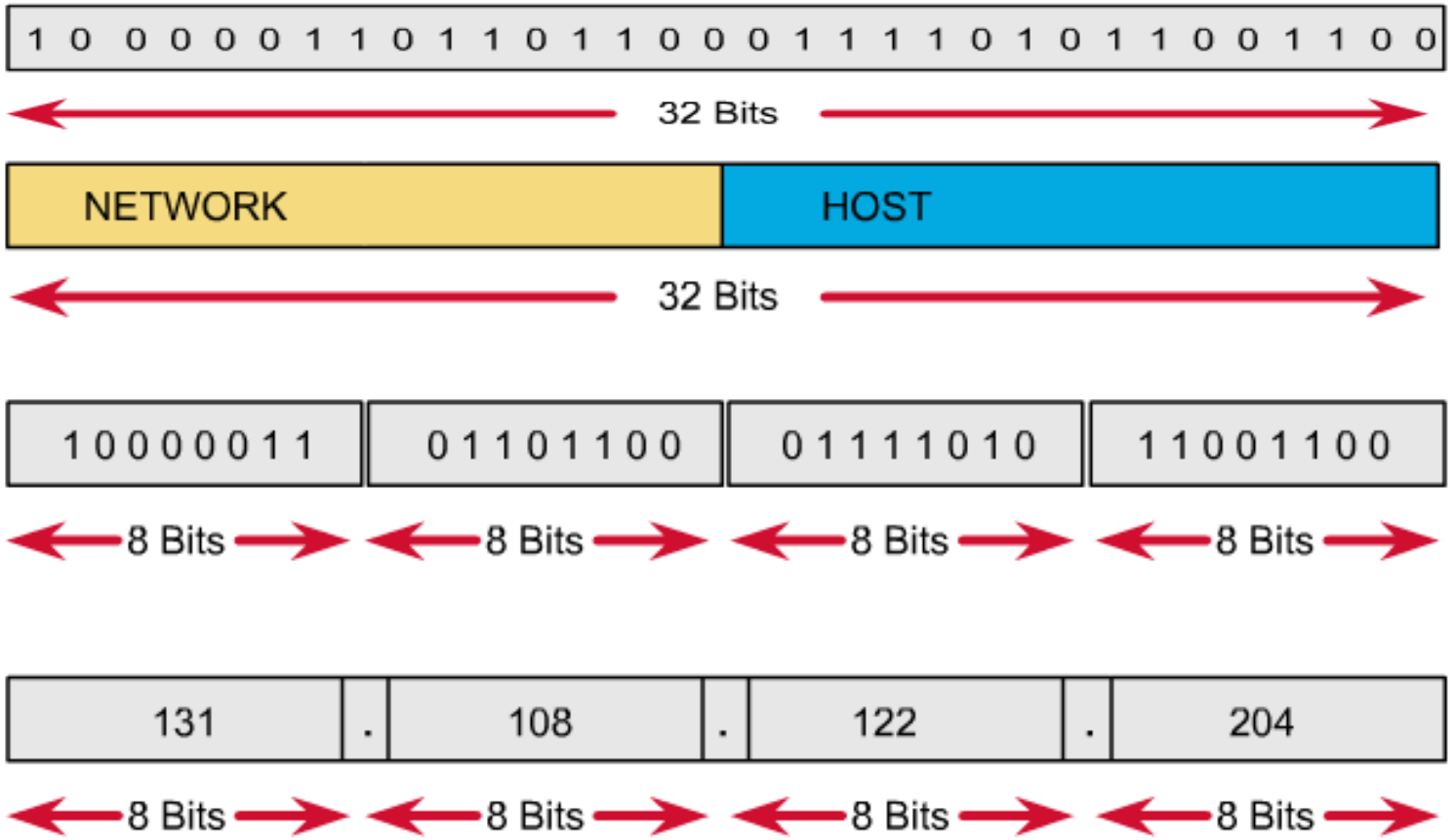
IP Address & Subnetting

IP Address

Constructing a Network Addressing Scheme

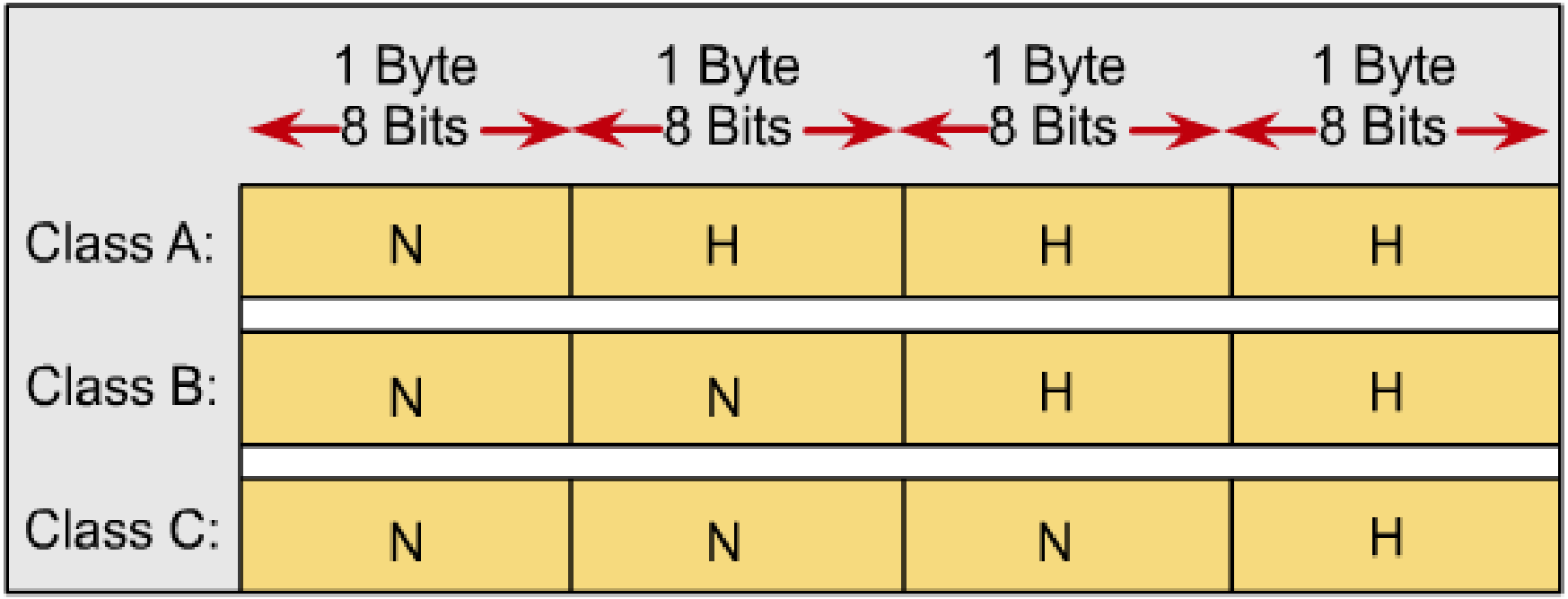


Constructing a Network Addressing Scheme (Cont.)

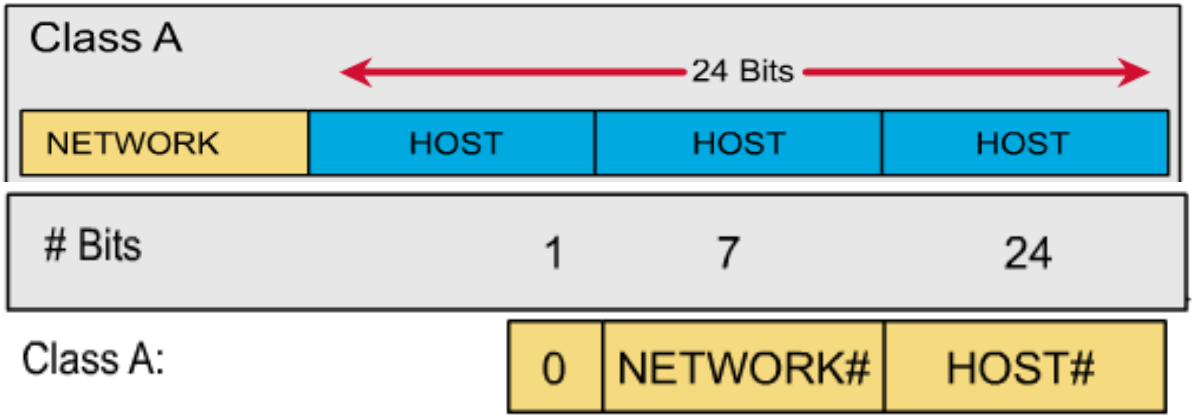
- Các bit phần mạng không được phép đồng thời bằng 0.
- Nếu tất cả bit phần host = 0 ->địa chỉ mạng.
- Nếu tất cả bit phần host = 1 ->địa chỉ quảng bá (broadcast).
  - Direct
    - Ví dụ: 192.168.1.255
  - Local
    - Ví dụ: 255.255.255.255



IP Address Classes: The First Octet

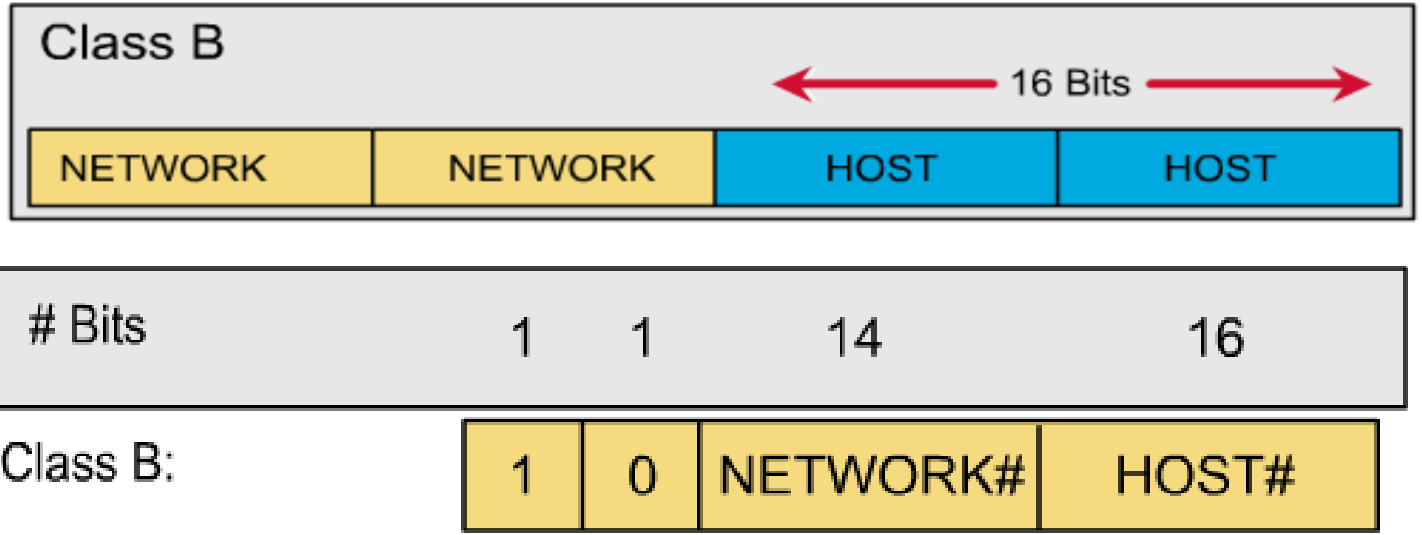


CLASS A



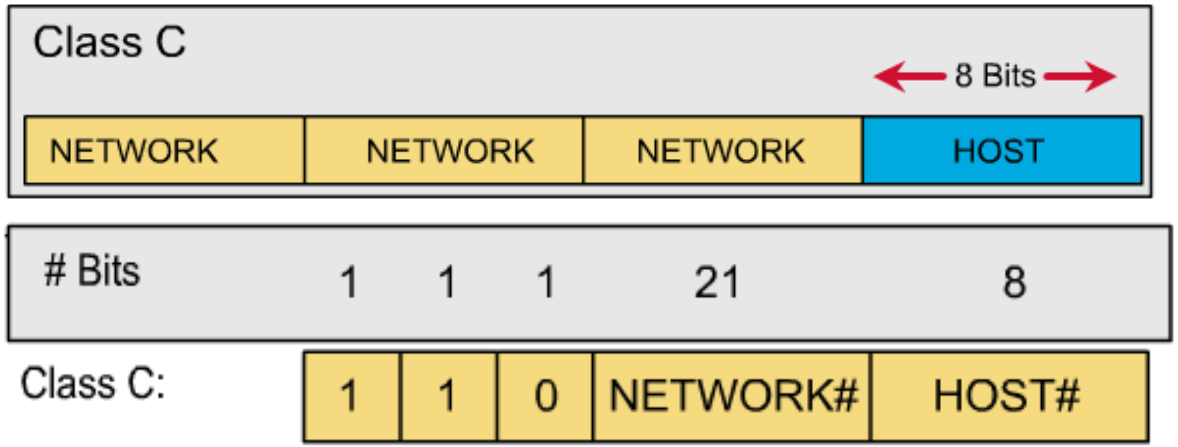
- Địa chỉ mạng: 1.0.0.0 → 127.0.0.0
- Mạng 127.0.0.0 : loopback network.
- Địa chỉ mạng sử dụng được: 1.0.0.0 →126.0.0.0 (126 mạng).
- Phần host: 24 bits ⇨ mỗi mạng lớp A có 224 – 2 host.

CLASS B



- Địa chỉ mạng: 128.0.0.0 → 191.255.0.0
- Có tất cả 214 mạng trong lớp B.
- Phần host: 16 bits ⇨ mỗi mạng lớp B có 216 – 2 host.

CLASS C



- Địa chỉ mạng: 192.0.0.0 → 223.255.255.0
- Có tất cả 221 mạng trong lớp B.
- Phần host: 8 bits ⇨ mỗi mạng lớp B có 28 – 2 host. (254 hosts)

CLASS D

- Địa chỉ mạng: 224.0.0.0 → 239.255.255.255
- Địa chỉ multicast.
- Ví dụ:
  - 224.0.0.5 dùng cho OSPF
  - 224.0.0.9 dùng cho RIPv2



CLASS E

- Từ 240.0.0.0 trở đi
- Dự phòng



Private and Public IP Addresses

- Mạng LAN: Private.
- Mạng Internet: Public.
- Dải địa chỉ private (RFC 1918):
  - Lớp A: 10.x.x.x
  - Lớp B: 172.16.x.x → 172.31.x.x
  - Lớp C: 192.168.x.x
- NAT: chuyển đổi private <-> public.
- Ý nghĩa của địa chỉ private: bảo tồn địa chỉ IP public.



Exercise

Cho biết địa chỉ nào sau đây có thể dùng cho host:

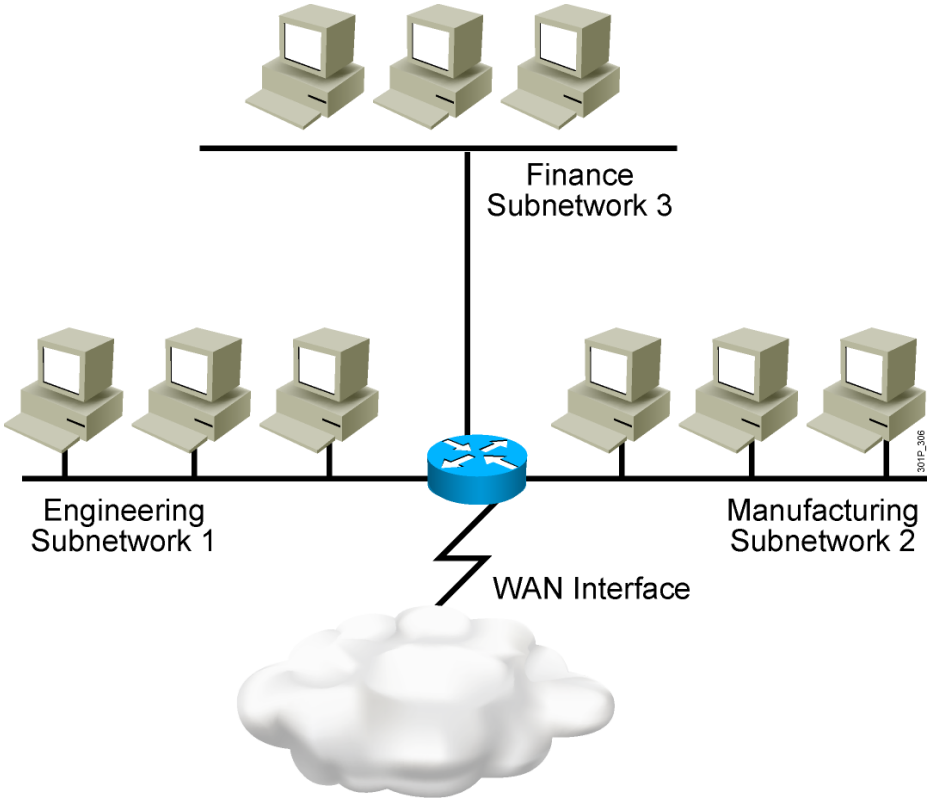
- 150.100.255.255
- 175.100.255.18
- 195.234.253.0
- 100.0.0.23
- 188.258.221.176
- 127.34.25.189
- 224.156.217.73



Subnetting

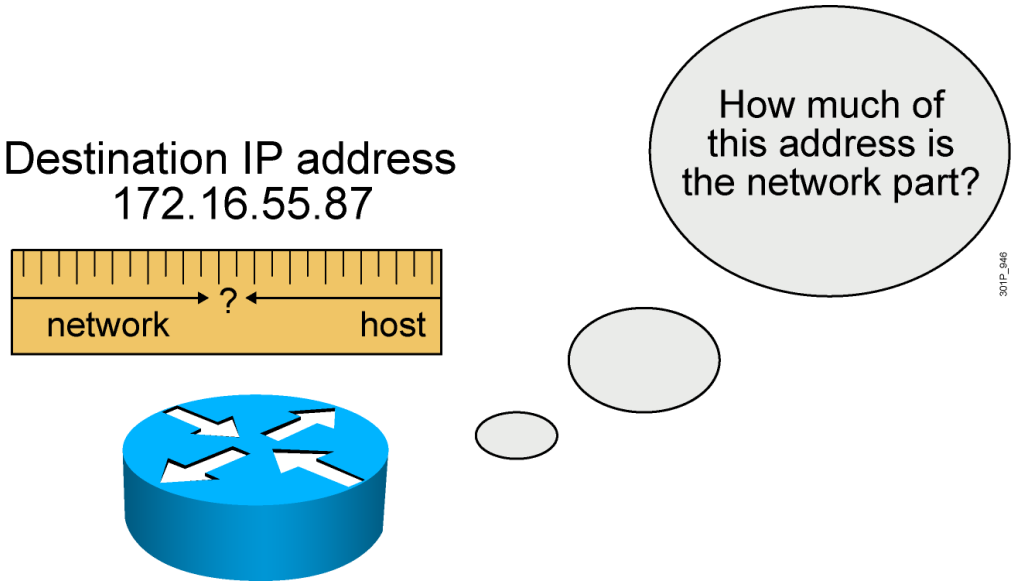
Subnetworks

- Smaller networks are easier to manage.
- Overall traffic is reduced.
- You can more easily apply network security policies.

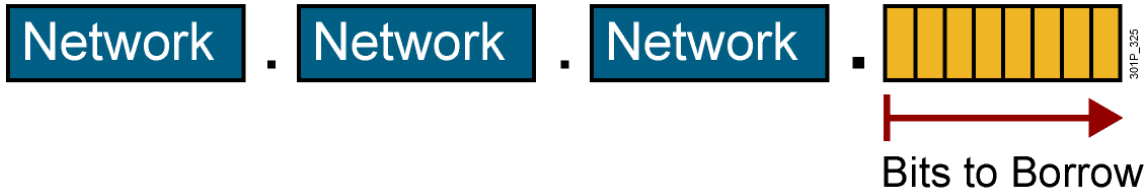


What a Subnet Mask Does

- Tells the router the number of bits to look at when routing
- Defines the number of bits that are significant
- Used as a measuring tool, not to hide anything

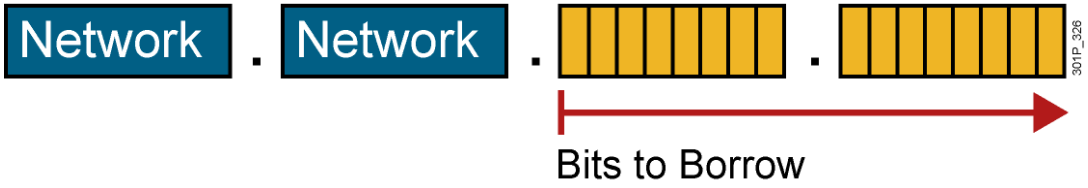


Possible Subnets and Hosts for a Class C Network



Number of Bits Borrowed (s)	Number of Subnets Possible (2^s)	Number of Bits Remaining in Host ID (8 - s = h)	Number of Hosts Possible Per Subnet ($2^h - 2$)
1	2	7	126
2	4	6	62
3	8	5	30
4	16	4	14
5	32	3	6
6	64	2	2
7	128	1	2

Possible Subnets and Hosts for a Class B Network

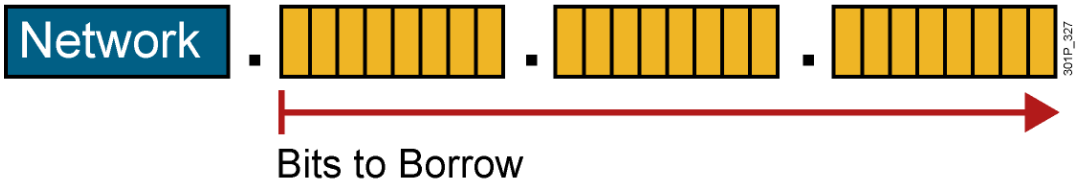


Number of Bits Borrowed (s)	Number of Subnets Possible (2^s)	Number of Bits Remaining in Host ID ($16 - s = h$)	Number of Hosts Possible Per Subnet ($2^h - 2$)
1	2	15	32,766
2	4	14	16,382
3	8	13	8,190
4	16	12	4,094
5	32	11	2,046
6	64	10	1,022
7	128	9	510
…	…	…	…

301P_304



Possible Subnets and Hosts for a Class A Network

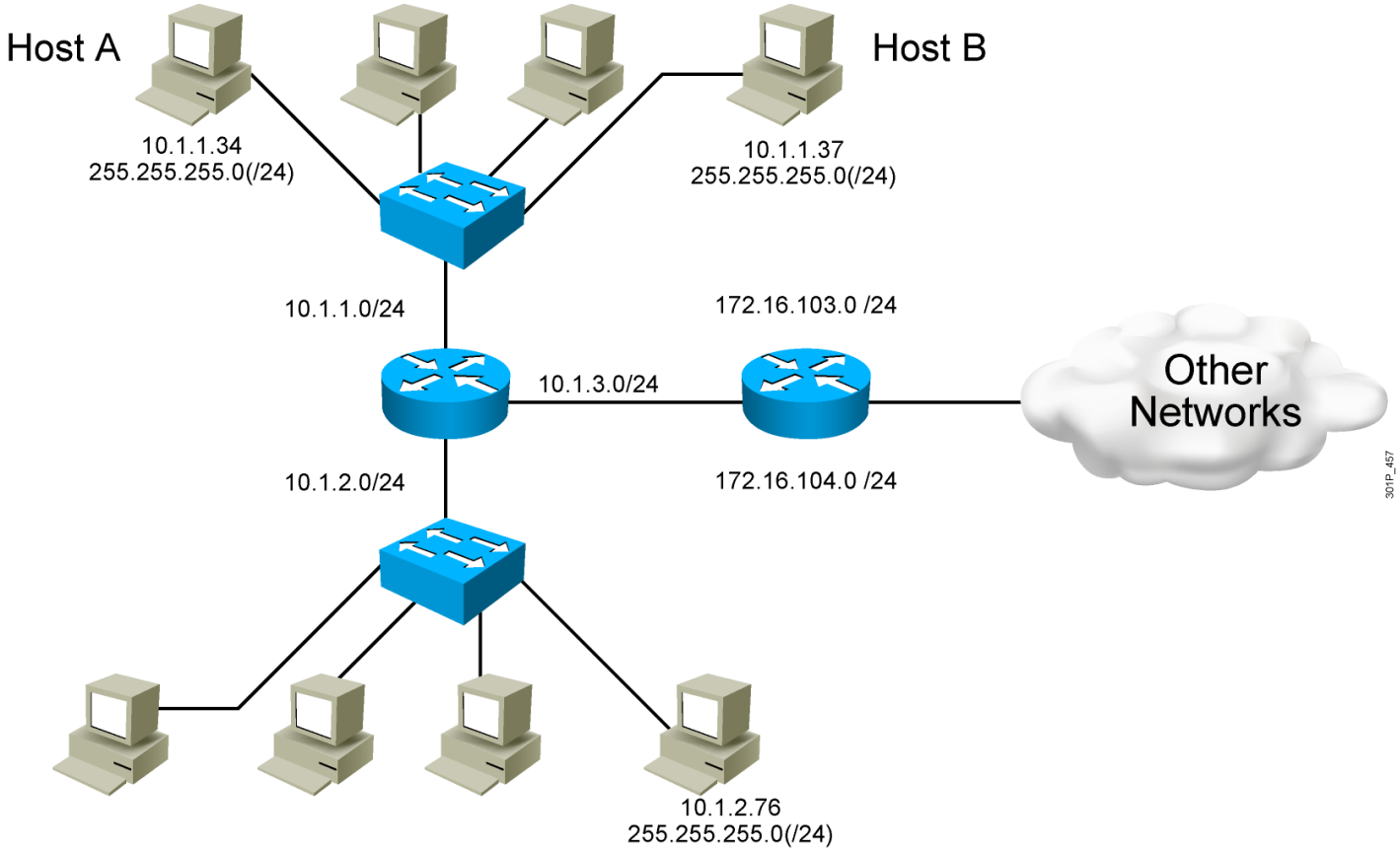


Number of Bits Borrowed (s)	Number of Subnets Possible (2^s)	Number of Bits Remaining in Host ID ($24 - s = h$)	Number of Hosts Possible Per Subnet ($2^h - 2$)
1	2	23	8,388,606
2	4	22	4,194,302
3	8	21	2,097,150
4	16	20	1,048,574
5	32	19	524,286
6	64	18	262,142
7	128	17	131,070
…	…	…	…

301P_303



End System Subnet Mask Operation



Octet Values of a Subnet Mask

128	64	32	16	8	4	2	1		
1	0	0	0	0	0	0	0	=	128
1	1	0	0	0	0	0	0	=	192
1	1	1	0	0	0	0	0	=	224
1	1	1	1	0	0	0	0	=	240
1	1	1	1	1	0	0	0	=	248
1	1	1	1	1	1	0	0	=	252
1	1	1	1	1	1	1	0	=	254
1	1	1	1	1	1	1	1	=	255

022P_164

- Subnet masks, like IP addresses, are represented in the dotted decimal format like 255.255.255.0



Default Subnet Masks

Example Class A address (decimal):	10.0.0.0
Example Class A address (binary):	00001010.00000000.00000000.00000000
Default Class A mask (binary):	11111111.00000000.00000000.00000000
Default Class A mask (decimal):	255.0.0.0
Default classful prefix length:	/8

Example Class B address (decimal):	172.16.0.0
Example Class B address (binary):	10010001.10101000.00000000.00000000
Default Class B mask (binary):	11111111.11111111.00000000.00000000
Default Class B mask (decimal):	255.255.0.0
Default classful prefix length:	/16

Example Class C address (decimal):	192.168.42.0
Example Class C address (binary):	11000000.10101000.00101010.00000000
Default Class C mask (binary):	11111111.11111111.11111111.00000000
Default Class C mask (decimal):	255.255.255.0
Default classful prefix length:	/24



Eight Easy Steps for Determining Subnet Addresses

IP Address: 192.168.221.37 Subnet Mask /29

Step	Description	Example
1.	Write the octet that is being split in binary.	Fourth octet: 00100101
2.	Write the mask or classful prefix length in binary.	Assigned mask: 255.255.255.248 (/29) Fourth octet: 11111000
3.	Draw a line to delineate the significant bits in the assigned IP address. Cross out the mask so you can view the significant bits in the IP address.	Split octet (binary): 00100101 Split mask (binary): 11111000

301P_301



Eight Easy Steps for Determining Subnet Addresses (Cont.)

Step	Description	Example
4.	Copy the significant bits four times.	00100 000 (network address) 00100 001 (first address in subnet) 00100 110 (last address in subnet) 00100 111 (broadcast address)?
5.	In the first line, define the network address by placing all zeros in the significant bits.	
6.	In the last line, define the broadcast address by placing all ones in the significant bits.	
7.	In the middle lines, define the first and last host number.	
8.	Increment the subnet bits by one.	00101 000 (next subnet)

Completed Subnet Addresses

Network address: 192.168.221.32
Subnet mask: 255.255.255.248
First subnet: 192.168.221.32
First host address: 192.168.221.33
Last host address: 192.168.221.38
Broadcast address: 192.168.221.39
Next subnet: 192.168.221.40

301P_936



Example: Applying a Subnet Mask for a Class C Address

IP Address 192.168.5.139 Subnet Mask 255.255.255.224

IP Address	192	168	5	139	
IP Address	11000000	10101000	00000101	10001011	
Subnet Mask	11111111	11111111	11111111	11100000	/27
Subnetwork	11000000	10101000	00000101	10000000	
Subnetwork	192	168	5	128	
First Host	192	168	5	10000001=129	
Last Host	192	168	5	10011110=158	
Directed Broadcast	192	168	5	10011111=159	
Next Subnet	192	168	5	10100000=160	

301P_168



Example: Applying a Subnet Mask for a Class B Address

IP Address 172.16.139.46 Subnet Mask /20

IP Address	172	16	139	46	
IP Address	10101100	00010000	10001011	00101110	
Subnet Mask	11111111	11111111	11110000	00000000	/20
Subnetwork	10101100	00010000	10000000	00000000	
Subnetwork	172	16	128	0	
First Host	172	16	10000000	00000001=128.1	
Last Host	172	16	10001111	11111110=143.254	
Directed Broadcast	172	16	10001111	11111111=143.255	
Next Subnet	172	16	10010000	00000000=144.0	

301P_169



Example: Applying a Subnet Mask for a Class A Address

IP Address 10.172.16.211 Subnet Mask /18

IP Address	10	172	16	211	
IP Address	00001010	10101100	00010000	11010011	
Subnet Mask	11111111	11111111	11000000	00000000	/18
Subnetwork	00001010	10101100	00000000	00000000	
Subnetwork	10	172	0	0	
First Host	10	172	00000000	00000001=0.1	
Last Host	10	172	00111111	11111110=63.254	
Directed Broadcast	10	172	00111111	11111111=63.255	
Next Subnet	10	172	01000000	00000000=64.0	

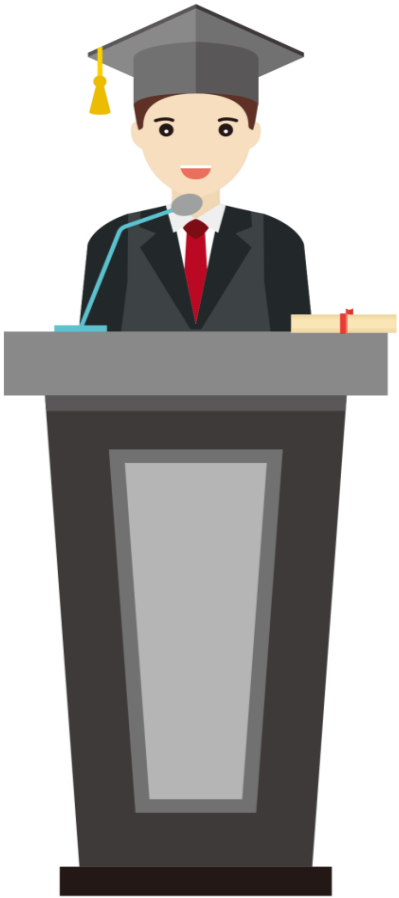
30IP_170



Summary

- IP network addresses consist of two parts: the network ID and the host ID.
- Networks, particularly large networks, are often divided into smaller subnetworks, or subnets. Subnets can improve network performance and control.
- A subnet address extends the network portion, and is created by borrowing bits from the original host portion and designating them as the subnet field.
- Determining the optimal number of subnets and hosts depends on the type of network and the number of host addresses required.
- The algorithm for computing a number of subnets is 2s, where s is the number of subnet bits.





THANK YOU!

